Flow Battery Reproducibility Research Committee (FBR-RC)

27th August 2026 – Kick-off meeting
15:00-16:00 (UTC+1)

Attendees:
Josh Bailey, Ed Brightman, Michael Gamble, Tommy George, Thomas Goodwin, Derek Hall, Maxime van der Heijden, William Johnson, Emily Kerr, Veryansen Lienuardy, Marco Cecchetti, Eleonora Natale, Tam Nguyen, Christian Stolze, Marina Tabuyo Martinez, Thomas Wakelin, Robin Wessling, Emilio Avellone, Andrea Trovó, David Reber

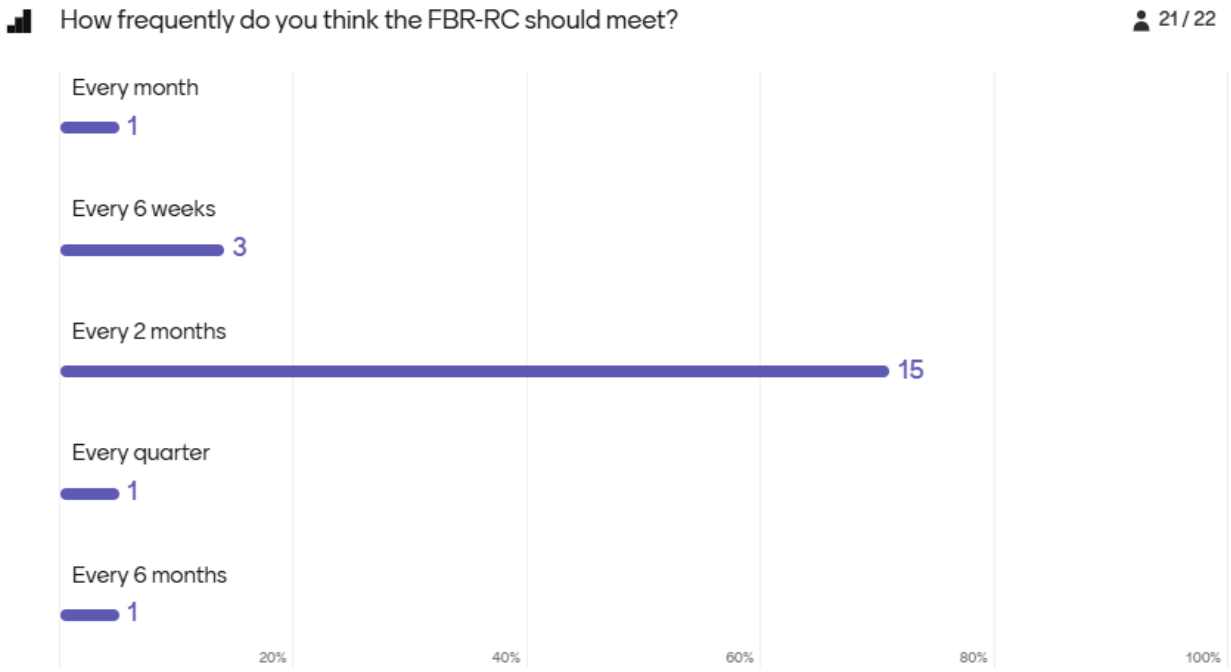
Could not attend:
Fikile Brushett, Johan Hjelm, Jose Rochin, Anthony Kucernak

MINUTES
Please note these minutes were taken by the chair once it was highlighted that minuting meetings was high on the priority list for committee ways of working. The minutes are therefore incomplete and comments unattributed. It is suggested that the Co-Chair or another nominated party take minutes at subsequent meetings.

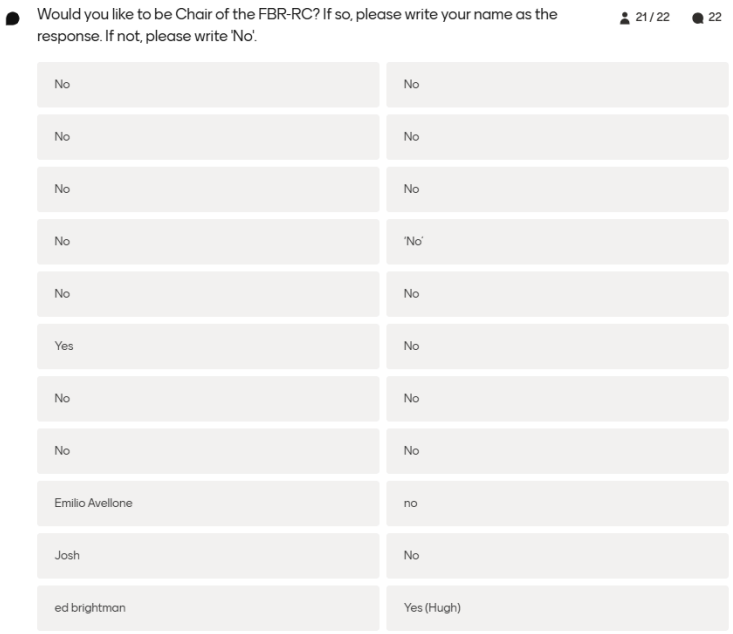
An introduction was given to kick-off the Research Committee, highlighting it as a space primarily to discuss the research elements of the on-going flow battery reproducibility efforts.

A sub-committee structure may be proposed in the next meeting to focus efforts.

A Mentimeter poll was taken by ~all participants. The questions and responses are below:

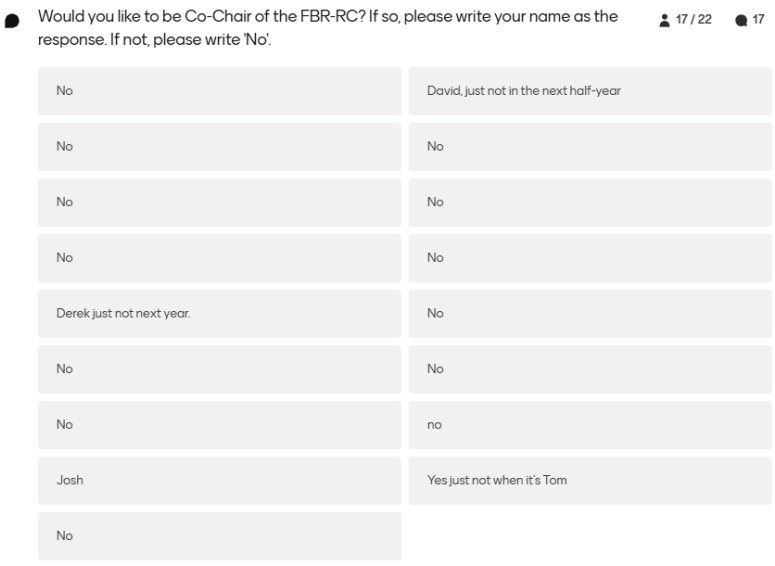


A two-month meeting cycle is adopted.



The first ‘Yes’ was Thomas Wakelin. Emilio Avellone, Josh Bailey, Ed Brightman and Hugh O’Connor also proposed themselves.

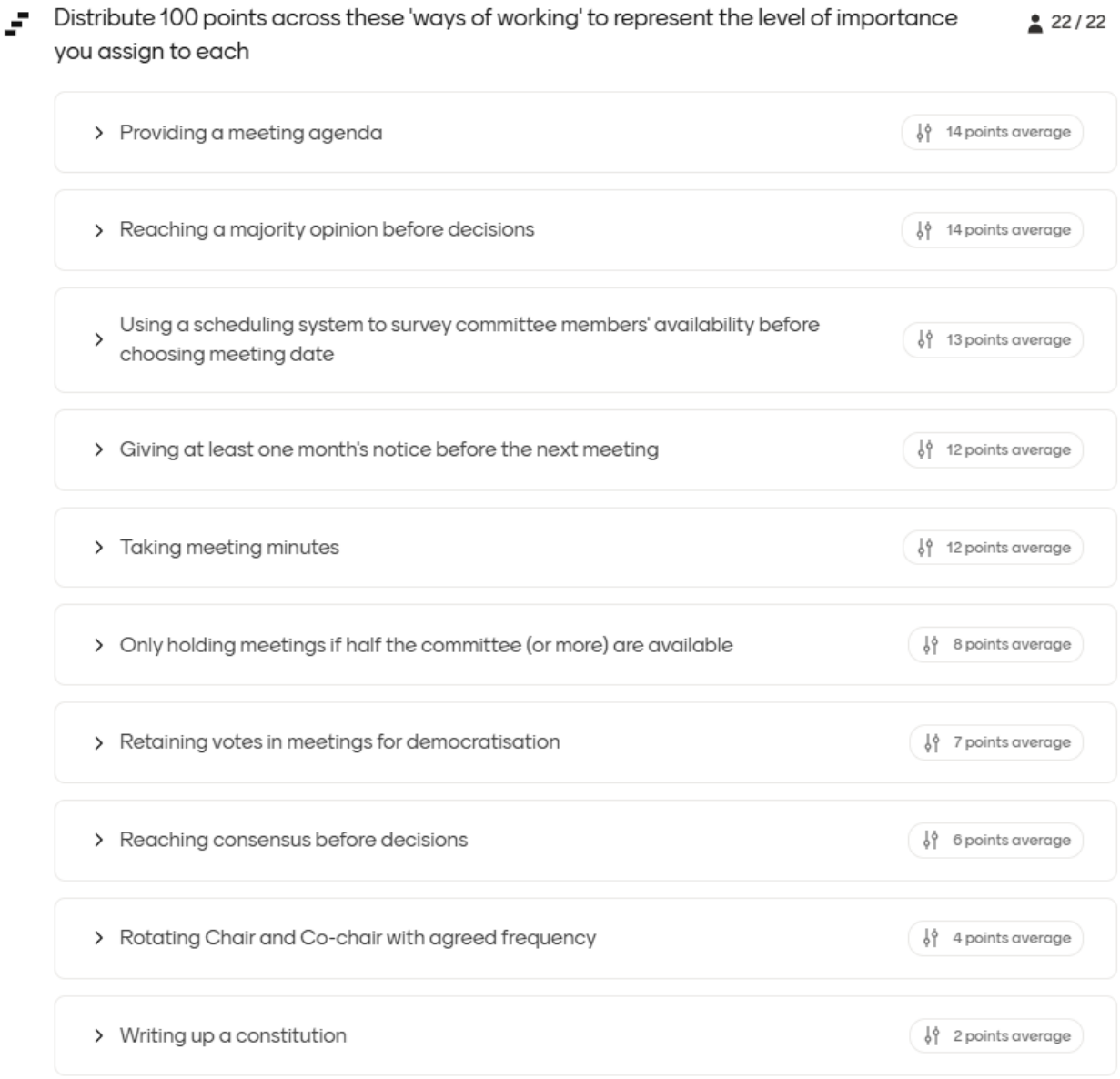
The Chair for the next session will be **Thomas Wakelin**. This next session, the second FBR-RC meeting, is to be held in ~2 months’ time.



David Reber, Derek Hall, Thomas Wakelin, and Josh Bailey proposed themselves as Co-Chairs. Given the potential Chairs and time stipulations for Co-Chairs, the following order is proposed:

First 6 months:	Chair: Thomas Wakelin;	Co-Chair: Derek Hall(/Josh Bailey*)
Second 6 months:	Chair: Emilio Avellone;	Co-Chair: David Reber
Third 6 months:	Chair: Ed Brightman;	Co-Chair: Josh Bailey
Fourth 6 months:	Chair: Hugh O’Connor;	Co-Chair: Thomas Wakelin
Fifth 6 months:	Chair: Josh Bailey;	Co-Chair: Derek Hall

**If Derek Hall needs to step down as Co-Chair next calendar year, rather than next academic year*



Greatest importance given to:

- Providing a meeting agenda – an agenda will be circulated before and presented during each meeting.
- Reaching a majority opinion before each decision – democratic voting and achieving majority, rather than full consensus, will be sought.
- Using a scheduling system for members' availability – see next point
- Give at least one month's notice before next week – 5 weeks prior to next meeting, poll will be sent out, giving members a week to fill in. 4 weeks prior to next meeting, meeting invitation circulated. 'Accept'/'Tentative'/'Decline' checked week before.*
- Taking meeting minutes - *it is suggested the Co-Chair or another nominated party at the meeting take minutes.*
- Meetings will be held as long as 5 or more participants are potentially available*, since the '50% threshold' scored only a mid-level response.

Update on Phase 2a:

PX	Major issue	Erroneous data	Other issues	On Dashboard?	Action may be needed?	Issue identified	Action needed
P02	-	Pol, FR 3.4, R1R2R3	-	Y	Y	Later current steps in POL hit voltage cut-offs	Remedy with updated processing
P03	-	CD, R1	-	Y	Y	Issue with charge-discharge raw data for R1	Remedy with updated processing
P05	-	-	-	Y	N	-	None
P06	1 Repeat	-	-	N	Y	Dataset not usable	Repeat attempts to contact participant failed
P07	-	Pol, FR 25.5, R2	-	Y	Y	Processing issue resolved	None
P08	-	CD, R1	Positive POL only	Y	Y	Variable time intervals identified	Remedy with updated processing
P09	-	-	-	Y	N	-	None
P11	-	Pol, R1	EIS, R1	Y	Y	Processing issue resolved	None
P12	-	CD, R3	Pol, FR 1.3, R1R2R3	Y	Y	Later current steps in POL hit voltage cut-offs	Remedy with updated processing
P13	-	Pol, R1R2R3	Low discharge capacity?	Y	Y	Processing issue resolved	None
P14	-	-	-	Y	N	-	None
P15	-	-	-	Y	N	-	None
P16	-	Pol, FR 78.0, R3	High EU?	Y	Y	Processing issue resolved	None
P18	-	Pol, FR 3.3, R1R2	-	Y	Y	Later current steps in POL hit voltage cut-offs	Remedy with updated processing
P19	1 Repeat	-	Low EU	Y	Y	Found other two repeats	None
P20	-	-	-	Y	N	-	None
P21	1 Repeat	Pol, FR 5.0, R1	Low discharge capacity?	Partially	Y	Incorrect procedure followed	2/3 new cell repeats complete; waiting on third
P22	-	CD, R1R2R3	Low discharge capacity?	Y	Y	Issues with cell	0/3 new cell repeats complete; waiting on all three
P24	1 Repeat	CD R2R3, Pol R2R3 All FR	-	N	Y	Issues with procedure	0/3 new cell repeats complete; waiting on all three
P25	-	-	High discharge capacity?	Y	Y	Issue with electrolyte molarity in first 2 cell repeats	0/3 new cell repeats complete; waiting on all three
P26	-	Pol, FR 12.5, R1	High discharge capacity?	Y	Y	Polarisation data incomplete for 1 cell repeat	0/1 new cell repeats complete; waiting on one
P28	-	-	-	Y	N	-	None
P29	-	CD, R2	-	Y	Y	Oscillation in CD raw data	Remedy with updated processing
P30	-	Pol, FR 78.9, R3	-	Y	Y	Issue with first set of repeats; provided second set	None
P31	-	-	-	Y	N	-	None
P32	-	Pol, FR 0.8 2.0, R1R2R3	-	Y	Y	Later current steps in POL hit voltage cut-offs	Remedy with updated processing
P33	Wrong volume	-	-	Y	Y	Issue with electrolyte volumes; ran new repeats	None
P34	-	-	-	Y	N	-	None
P35	-	Pol, FR 21.5, R2 (one bad data point)	-	Y	N	-	None
P36	-	CD, R1R2R3 (error in data cut-offs)	CD, R3	Y	Y	Processing issue resolved	None
P37	-	Pol, FR 3.4, R1R2R3	Low EU?	Y	Y	Later current steps in POL hit voltage cut-offs	Remedy with updated processing
P39	-	Pol, R1R2R2 (missing data)	-	Y	N	-	None
P40	-	CD, R1 (cycle 9 decay)	-	Y	Y	Sufficient data before capacity fade in late cycle	None
P41	-	Pol, R1R2R3, all FRs	-	Y	Y	Issues with cell	0/3 new cell repeats complete; waiting on all three
P43	-	Pol, FR 1.3, R1R2R3	-	Y	Y	Issue with polarissation protocol	0/3 new cell repeats complete; waiting on all three
P44	-	-	-	Y	N	-	None
P45	-	-	-	Y	N	-	None
P46		Pol, FR 10.1, R1	EIS, FR 55.7, R2	Y	Y	Issue with data collection	0/3 new cell repeats complete; waiting on all three

38 datasets in total:

- 21 datasets ready for analysis
- 8 datasets can be remedied with updated processing (P02,03,08,12,18,29,32,37)
- 8 datasets: waiting on 1-3 new cell repeats (P21,22,24,25,26,41,43,46)
- 1 dataset is not usable (P06)

8 datasets can be ‘cleaned’ with minor changes to current processing procedures.
8 datasets are pending: most, if not all, of these participants likely to return data by end of September/beginning of October.

Data publication:
An opportunity to publish with Frontiers was presented.

Publisher: Frontiers
Journal: Frontiers in Batteries and Electrochemistry
Research Topic: Advancing Open Science in Batteries and Electrochemistry: A Collection of FAIR² Data Articles
Deadline: 17th January 2027

The following feedback was provided:

-A question arose around what benefit there is in doing this, rather than directly publishing on Zenodo
-Potential benefits may arise from: visualisation of the data; more context given in a data article than a simple raw data drop; the data may have to adhere more closely to FAIR principles than result if done directly; the upload and AI data steward tools may streamline the process; the result may look more appealing; the journal may make more of a concerted effort to advertise thus drawing in a bigger audience.
-A member had previously had a negative experience with the publisher, potentially indicative of predatory practices (around APCs).

Clear decision: the community wishes to publish the data before the full article (or articles) come out with the analysis thereof.

Three parallel options being pursued:
1) Explore Frontiers route (with Data Steward Clara)
2) Investigate direct publication on Zenodo
3) Alternative data article publisher

The following people have volunteered to pursue these:
1) Josh Bailey, Christian Stolze, David Reber, Marco Cecchetti, H. O'Connor, Tom Goodwin*
2) Christian Stolze
3) David Reber, Derek Hall, Tommy George, Tam Nguyen, Hugh O'Connor

**Josh Bailey, Hugh O'Connor and Tom Goodwin to send to Christian Stolze, David Reber and Marco Cecchetti, two clean anonymised datasets, their corresponding metadatasets, and a README version of the experimental ask as model data to use in interfacing with AI data steward.*

Data Analysis

The results of the poll from the previous community call were presented and the broad categories of 'analysis' were briefly outlined.

Yes, I'd like to help organise a group of community members to analyse the data	Derek Hall	Graham Smith	Tom Wakelin	Ed Saunders	
	Josh Bailey	Thomas Goodwin	Hugh O'Connor	Fikile Brushett	
Yes, I'd like to help analyse the data as part of a group of members	Alex Quinn	William Johnson	Sofía Edgar	Edoardo Macchi	
	Robin Wessling	Emilio Avellone	Emily Kerr	Anthony Kucernak	
	Camilo Zuleta	Christian Stolze	Veryansen Lienuardy	Tam Nguyen	Daniel McCormack
	Sonia Khalghollah	Andrea Trovó	Hugh O'Connor	Eleonora Natale	Tom Hinson

	If you are interested, what kind of data analysis would you like to do? (Select all that apply)				If you are interested, which parts of the dataset are you interested in analysing? (Select all that apply)		
Derek Hall	ML/AI	Fundamental			CD	Pol	EIS
Graham Smith	ML/AI	Fundamental	Statistical		CD	Pol	EIS
Tom Wakelin	ML/AI	Fundamental	Statistical		CD	Pol	EIS
Ed Saunders	ML/AI	Fundamental					EIS
Josh Bailey	ML/AI	Fundamental	Statistical	Comparisons	CD	Pol	EIS
Alex Quinn		Fundamental	Statistical		CD	Pol	EIS
William Johnson			Statistical	Comparisons		Pol	
Sofía Edgar			Statistical	Comparisons	CD		
Marco Cecchetti		Fundamental		Comparisons	CD	Pol	EIS
Camilo Zuleta	ML/AI	Fundamental	Statistical	Comparisons	CD	Pol	EIS
Christian Stolze			Statistical	Comparisons	CD	Pol	EIS
Veryansen Lienuardy			Statistical		CD		
Tam Nguyen		Fundamental		Comparisons	CD		
Daniel McCormack		Fundamental		Comparisons	CD		EIS
Hugh O'Connor	ML/AI	Fundamental	Statistical	Comparisons	CD	Pol	EIS
Eleonora Natale				Comparisons		Pol	
Tom Hinson			Statistical		CD		
Robin Wessling	ML/AI	Fundamental	Statistical	Comparisons	CD		
Emilio Avellone		Fundamental					EIS
Emily Kerr	ML/AI			Comparisons	CD		
Anthony Kucernak		Fundamental			CD	Pol	EIS
Andrea Trovó				Comparisons	CD	Pol	EIS
Sonia Khalghollah			Statistical		CD	Pol	
	ML/AI	**Fundamental**	**Statistical**	**Comparisons**	**CD**	**Pol**	**EIS**
	9	**14**	**13**	**13**	**19**	**14**	**14**

Majority opinion supported the idea of creating sub-groups for each of the four categories, matching preferences as best as possible, whilst attempting to balance numbers.

General approach:

- Sub-group meets and makes initial assessment, reports back in main committee meeting.
- There may be a need to iron out what is covered by what, avoiding overlap as best as possible.
- Identifying what is covered by the sub-groups for 'ML/AI', 'Statistics' and 'Comparisons' may be contingent upon preliminary 'Fundamental' analysis outputs.
- Individual #channels in Discord to be used to keep conversations and documents well classified.

Josh Bailey to form 4 sub-groups in coming weeks, based on above allocations, and kickstart discussions on: what research questions are being asked; what tools are available; what is in and what is beyond scope.